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mQTL Meta-Analysis: PEG + PPMI

[Code ▾](#)

Your Name

2026-08-12

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```
suppressPackageStartupMessages({  
  library(tidyverse)  
  library(data.table)  
  library(meta)          # r-meta: metagen() for inverse-variance meta-analysis  
  library(ggplot2)  
  library(patchwork)  
  library(scales)  
})
```

1. Configuration

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```

# — Paths to cis mQTL result files —
—
RESULTS <- list(
  peg1_cases      = "../results/PEG_MQTL/mqtl/peg1_cases/cis_mqtls.txt",
  peg1_controls   = "../results/PEG_MQTL/mqtl/peg1_controls/cis_mqtls.txt",
  peg2_cases      = "../results/PEG_MQTL/mqtl/peg2_cases/cis_mqtls.txt",
  ppmi_cases      = "../results/PPMI_MQTL/mqtl/ppmi_cases/cis_mqtls.txt",
  ppmi_controls   = "../results/PPMI_MQTL/mqtl/ppmi_controls/cis_mqtls.txt"
)

# Row cap per cohort file. The per-cohort cis thresholds are deliberately
# liberal (1e-2 for PEG_MQTL, 1e-3 for PPMI_MQTL) so that enough pairs survive
# for the meta-analysis to combine evidence across cohorts – which makes the
# raw files far too large to read whole on this machine.
#
# This cap is NOT the same one the overlap report applies: that one bounds the
# PEG_METAB trans metQTL files, which this report never reads. Nothing caps
# these files at write time, so this is the only guard on the cis path.
#
# Note this is a real analytic trade-off, not just a memory guard: truncation
# happens before cohorts are pooled, so a pair that would have reached
# significance only by combining weak-but-concordant evidence across cohorts
# can be cut. The load step warns when a file actually hits the cap.
#
# Raised 10,000,000 -> 15,000,000 on 2026-08-12. At 10M this truncated the two
# largest cohorts (peg1_cases 10,266,866 rows; ppmi_cases 11,230,706), dropping
# pairs before they could be pooled. Nothing on disk now exceeds 11.3M, so 15
# reads every cohort whole – and because a cap only bites on files larger than
# itself, 15M costs exactly what 12M would today while leaving room for growth.
#
# Memory: the 10M run peaked at 18.5 GB of 31 GB (~450 MB per million rows via
# fread). The extra ~1.5M rows should put it near 19 GB. This report, not the
# overlap report, is the memory-bound one – the overlap report peaked at 5.4
# GB
# because readr produces far leaner structures than fread. If a future runOOMs
# here, lower this before touching MAX_TRANS_ROWS.
MAX_CIS_ROWS <- 15000000

# PEG_MQTL and PPMI_MQTL now write gzipped output (matrixeqtl.gzip_output in

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```

# their config.yaml). Prefer the .gz when it exists so this report works both
# before and after those pipelines are re-run, without editing paths twice.
resolve_path <- function(p) if (file.exists(paste0(p, ".gz"))) paste0(p, ".gz") else p
RESULTS <- lapply(RESULTS, resolve_path)

# Meta-analysis groupings
# Cases:      PEG1 x PPMI two-way – comparable sample sizes to controls
#            PEG2 excluded from primary; used as sensitivity analysis below
# Controls: PEG1 x PPMI two-way
# Rationale: 3-way PEG1 n PEG2 n PPMI yields only ~2 pairs;
#            PEG1 n PPMI yields ~1032 pairs which is tractable.
META_GROUPS <- list(
  cases      = c("peg1_cases", "ppmi_cases"),
  controls   = c("peg1_controls", "ppmi_controls")
)

# Sensitivity: 3-way intersection including PEG2 cases
SENSITIVITY_GROUPS <- list(
  cases_peg2_sensitivity = c("peg1_cases", "peg2_cases", "ppmi_cases")
)

# Output directory
META_DIR <- "../results/META_MQTL/meta"
dir.create(META_DIR, recursive=TRUE, showWarnings=FALSE)
dir.create("../results/META_MQTL/qc", recursive=TRUE, showWarnings=FALSE)

# Labels and colors
DATASET_LABELS <- c(
  peg1_cases      = "PEG1 Cases",
  peg1_controls   = "PEG1 Controls",
  peg2_cases      = "PEG2 Cases",
  ppmi_cases      = "PPMI Cases",
  ppmi_controls   = "PPMI Controls"
)

DATASET_COLORS <- c(
  peg1_cases      = "#E41A1C",
  peg1_controls   = "#377EB8",
  peg2_cases      = "#FF7F00",
  ppmi_cases      = "#4DAF4A",
  ppmi_controls   = "#984EA3"
)

```

2. Load mQTL Results

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```

load_mqtl <- function(path, dataset_name) {
  message(sprintf("Loading dataset %s", dataset_name))
  if (!file.exists(path)) {
    warning(sprintf("File not found: %s", path))
    return(NULL)
  }
  # MatrixEQTL writes p-value ascending when it computes FDR
  # (noFDRsaveMemory = FALSE), so capping keeps the most significant rows and
  # discards only the tail. See MAX_CIS_ROWS above for the trade-off.
  #
  # Read via `zcat | head` rather than fread(nrows=) on the gzip directly:
  # fread decompresses the *entire* file before honouring nrows (measured: re
ad
  # time scales with total file size, 1.8s -> 5.4s for a 4x larger file at th
e
  # same cap), which on a multi-GB cis file means expanding every row we are
  # about to discard. Piping through head stops at the cap – constant time an
d
  # the lowest peak memory of the options tested.
  if (grepl("\\.gz$", path)) {
    dat <- fread(cmd = sprintf("zcat %s | head -n %.0f",
                                shQuote(path), MAX_CIS_ROWS + 1),
                  data.table = FALSE)
  } else {
    dat <- fread(path, data.table = FALSE, nrows = MAX_CIS_ROWS)
  }

  # A failed decompress or a bad path surfaces as an empty read rather than a
n
  # error, which would otherwise sail past the file.exists() check above and
  # silently drop a whole cohort from the meta-analysis.
  if (nrow(dat) == 0) {
    warning(sprintf("%s: read 0 rows from %s – decompression or path proble
m?",
                    dataset_name, path))
    return(NULL)
  }

  # Reported with cat(), not warning(): this document sets warning = FALSE in
  # its global chunk options, so a warning() here is swallowed outright and t
he
  # truncation would go unrecorded. Truncation changes which pairs can reach
  # significance, so it belongs in the rendered output, not the console.
  if (nrow(dat) == MAX_CIS_ROWS) {
    pcol <- intersect(c("p-value", "p.value", "pvalue"), colnames(dat))
    worst_p <- if (length(pcol) > 0) max(dat[[pcol[1]]], na.rm=TRUE) else NA_
real_
    cat(sprintf(

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paste0(" *** %s: HIT THE %s-ROW CAP – least significant rows dropped
",
      "before meta-analysis (worst p retained: %.3g). Pairs that would
",
      "only reach significance by pooling across cohorts may be lost.
",
      "Raise MAX_CIS_ROWS if RAM allows. ***\n"),
dataset_name, format(MAX_CIS_ROWS, big.mark=","), worst_p))
}

# Normalise column names – MatrixEQTL output varies between pipelines:
# PEG uses: SNP gene beta t-stat p-value FDR
# PPMI uses: snps gene beta t-stat p-value FDR
col_map <- c(
  "SNP"      = "snp_id",
  "snps"     = "snp_id",
  "gene"     = "cpq_id",
  "p-value"  = "pvalue",
  "p.value"  = "pvalue",
  "t-stat"   = "tstat",
  "t.stat"   = "tstat",
  "FDR"      = "fdr_dataset"
)
matches <- col_map[names(col_map) %in% colnames(dat)]
if (length(matches) > 0)
  setnames(dat, old=names(matches), new=unname(matches))

required <- c("snp_id", "cpq_id", "beta", "pvalue", "tstat")
missing <- setdiff(required, colnames(dat))
if (length(missing) > 0) {
  warning(sprintf(
    "%s: missing columns after rename: %s\nActual columns: %s",
    dataset_name, paste(missing, collapse=" ", ),
    paste(colnames(dat), collapse=" ", )
  ))
  return(NULL)
}

# — Normalise SNP IDs to chr:pos so pair_id matches across cohorts —
—
# PEG SNP IDs are already chr:pos (e.g. "1:1446221") from imputation.
# PPMI SNP IDs are chr:pos:ref:alt (e.g. "1:1365759:G:A") from WGS.
# Stripping any trailing :REF:ALT normalises both to a common chr:pos key
# without needing a lookup join, which avoids brittleness if the position
# files and mQTL output ever drift out of sync.
dat$snp_chrpos <- sub("^( [0-9XYM]+:[0-9]+):[ACGT]+:[ACGT]+$", "\\1", dat$snp_id)

n_stripped <- sum(dat$snp_chrpos != dat$snp_id)

```

```

cat(sprintf("  %-20s: %s SNP IDs had allele suffix stripped (%s unchanged)
\n",
  dataset_name,
  format(n_stripped, big.mark=","),
  format(nrow(dat) - n_stripped, big.mark=",")))

dat <- dat %>%
  mutate(
    dataset = dataset_name,
    se       = abs(beta / tstat),
    pair_id = paste(snp_chrpos, cpq_id, sep="|")
  ) %>%
  filter(!is.na(se), se > 0, !is.na(beta), !is.na(pvalue))

cat(sprintf("  %-20s: %s associations\n",
  dataset_name, format(nrow(dat), big.mark=",")))
dat
}

cat("Loading cis mQTL results:\n")

```

```
## Loading cis mQTL results:
```

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```

message("Loading cis mQTL results:\n")
mqtl_list <- imap(RESULTS, function(path, nm) {
  load_mqtl(path, nm)
})

```

```
##   peg1_cases           : 0 SNP IDs had allele suffix stripped (10,266,866 u
nchanged)
##   peg1_cases           : 10,266,866 associations
```

```
##   peg1_controls        : 0 SNP IDs had allele suffix stripped (8,087,599 un
changed)
##   peg1_controls        : 8,087,599 associations
```

```
##   peg2_cases           : 0 SNP IDs had allele suffix stripped (9,099,593 un
changed)
##   peg2_cases           : 9,099,593 associations
```

```
## ppmi_cases : 0 SNP IDs had allele suffix stripped (11,230,706 unchanged)
## ppmi_cases : 11,230,706 associations
```

```
## ppmi_controls : 0 SNP IDs had allele suffix stripped (4,190,732 unchanged)
## ppmi_controls : 4,190,732 associations
```

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```
mqtL_list <- Filter(Negate(is.null), mqtL_list)

cat(sprintf("\nDatasets loaded: %d / %d\n",
            length(mqtL_list), length(RESULTS)))
```

```
##
## Datasets loaded: 5 / 5
```

3. P-value QC

3.1 Result Summary

Note on Lambda GC: Lambda GC is not reported here because MatrixEQTL outputs only associations passing the significance threshold ($pval_cis = 1e-3$ for PEG, $pval_cis = 1e-5$ for PPMI). Computing lambda on a filtered tail of the p-value distribution produces artificially inflated values that are meaningless. Lambda GC is only valid when computed on the complete null distribution. QQ plots in Section 3.2 provide valid visual QC for the significant associations we do have.

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```

summary_df <- imap_dfr(mqtl_list, function(dat, nm) {
  data.frame(
    dataset      = nm,
    label        = DATASET_LABELS[nm],
    n_assoc      = nrow(dat),
    n_fdr05      = if ("fdr_dataset" %in% colnames(dat))
                  sum(dat$fdr_dataset < 0.05, na.rm=TRUE) else NA,
    n_ple5       = sum(dat$pvalue < 1e-5, na.rm=TRUE),
    n_ple8       = sum(dat$pvalue < 1e-8, na.rm=TRUE),
    median_beta  = round(median(abs(dat$beta), na.rm=TRUE), 4),
    max_beta     = round(max(abs(dat$beta), na.rm=TRUE), 4),
    n_extreme    = sum(abs(dat$beta) > 1, na.rm=TRUE),
    stringsAsFactors = FALSE
  )
})

summary_df %>%
  mutate(
    n_assoc = format(n_assoc, big.mark=","),
    n_fdr05 = format(n_fdr05, big.mark=","),
    n_ple5  = format(n_ple5, big.mark=","),
    n_ple8  = format(n_ple8, big.mark=",")
  ) %>%
  select(label, n_assoc, n_fdr05, n_ple5, n_ple8,
         median_beta, max_beta, n_extreme) %>%
  knitr::kable(
    col.names = c("Dataset", "N assoc", "FDR<0.05", "p<1e-5", "p<1e-8",
                  "Median |β|", "Max |β|", "N |β|>1 (outliers)"),
    caption   = paste("Per-dataset cis-mQTL summary.",
                      "FDR computed by MatrixEQTL over all tested cis pairs",
                      "(noFDRsaveMemory=FALSE) – valid for within-dataset use.",
                      "Cross-cohort FDR (FDR_BH) is computed on meta-analysis",
                      "p-values in Section 5.")
  )

```

Per-dataset cis-mQTL summary. FDR computed by MatrixEQTL over all tested cis pairs (noFDRsaveMemory=FALSE) – valid for within-dataset use. Cross-cohort FDR (FDR_BH) is computed on meta-analysis p-values in Section 5.

	Dataset	N assoc	FDR<0.05	p<1e-5	p<1e-8	Median β	Max β	N β >1 (outliers)
peg1_cases	PEG1 Cases	10,266,866	14,853	51,616	2,419	0.0116	0.2499	0

	Dataset	N assoc	FDR<0.05	p<1e-5	p<1e-8	Median β	Max β	N β >1 (outliers)
peg1_controls	PEG1 Controls	8,087,599	914	16,969	301	0.0125	0.2820	0
peg2_cases	PEG2 Cases	9,099,593	1,391	31,172	380	0.0130	0.2979	0
ppmi_cases	PPMI Cases	11,230,706	8,583,873	4,749,259	2,365,291	0.0232	0.6372	0
ppmi_controls	PPMI Controls	4,190,732	1,548,936	1,022,012	354,943	0.0387	300.2187	48

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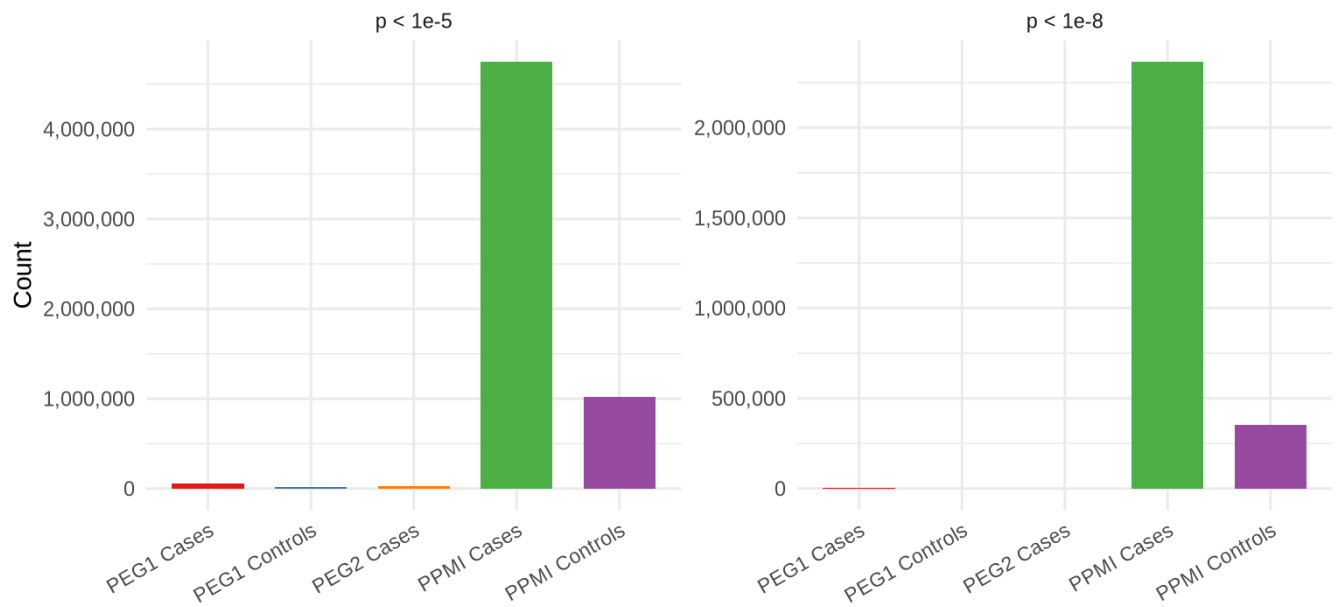
```
# — Flag extreme beta values —
#
# Methylation beta values should be bounded 0-1. Effect sizes (beta) in mQTL
# analysis represent change in methylation per unit allele dosage and should
# therefore be well within [-1, 1] for a properly scaled analysis.
# Values >> 1 indicate a scaling issue in the methylation or covariate matrix.
extreme_datasets <- summary_df %>% filter(max_beta > 1)
if (nrow(extreme_datasets) > 0) {
  cat("\n⚠ WARNING: Extreme beta values detected (| $\beta$ | > 1):\n")
  for (i in seq_len(nrow(extreme_datasets))) {
    cat(sprintf("  %-20s: max| $\beta$ | = %.2f, n outliers = %s\n",
                extreme_datasets$dataset[i],
                extreme_datasets$max_beta[i],
                format(extreme_datasets$n_extreme[i], big.mark=",")))
  }
  cat("\n  This most likely indicates a scaling issue in the PPMI methylation
\n\n")
  cat("  or covariate matrix. Possible causes:\n")
  cat("  - Methylation matrix not in beta [0,1] scale (e.g. M-values used)
\n")
  cat("  - A covariate column with large numeric range (e.g. age in years
\n")
  cat("    not standardised) creating numerical instability in MatrixEQTL
\n")
  cat("  - Batch variable included without dummy-coding\n")
  cat("  Recommend investigating PPMI results before proceeding with meta-analysis.\n")
}
```

```
##
## ⚠ WARNING: Extreme beta values detected ( $|\beta| > 1$ ):
## ppmi_controls      : max $|\beta|$  = 300.22, n outliers = 48
##
## This most likely indicates a scaling issue in the PPMI methylation
## or covariate matrix. Possible causes:
## - Methylation matrix not in beta [0,1] scale (e.g. M-values used)
## - A covariate column with large numeric range (e.g. age in years
##   not standardised) creating numerical instability in MatrixEQTL
## - Batch variable included without dummy-coding
## Recommend investigating PPMI results before proceeding with meta-analysis.
```

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```
# Bar chart of association counts at different p-value thresholds
summary_df %>%
  select(label, dataset, n_ple5, n_ple8) %>%
  pivot_longer(c(n_ple5, n_ple8),
               names_to="threshold", values_to="n") %>%
  mutate(threshold = factor(threshold,
                             levels = c("n_ple5", "n_ple8"),
                             labels = c("p < 1e-5", "p < 1e-8"))) %>%
  ggplot(aes(x=label, y=n, fill=dataset)) +
  geom_col(position="dodge", width=0.7) +
  scale_fill_manual(values=DATASET_COLORS, guide="none") +
  scale_y_continuous(labels=comma) +
  facet_wrap(~threshold, scales="free_y") +
  labs(title="Significant Associations per Dataset",
        x=NULL, y="Count") +
  theme_minimal(base_size=11) +
  theme(axis.text.x=element_text(angle=30, hjust=1))
```

Significant Associations per Dataset



3.2 QQ Plots

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```

# n_total: true number of tests the retained p-values were drawn from
# (e.g. all cis SNP-CpG pairs MatrixEQTL actually evaluated), when the
# input pvalues are only the smallest n of a much larger, left-truncated
# set. Order statistic i of n retained values is then the i-th smallest
# of n_total, not of n, so expected quantiles and the Beta CI ribbon
# must be computed against n_total. Defaults to n (i.e. assumes pvalues
# is the complete, untruncated set) when n_total is not supplied.
make_qq <- function(pvalues, title, color="steelblue", max_pts=1e6, n_total=N
ULL) {
  pvalues <- pvalues[!is.na(pvalues) & pvalues > 0 & pvalues <= 1]
  top <- pvalues <= 1e-4
  if (sum(!top) > max_pts)
    pvalues <- c(pvalues[top],
                  sample(pvalues[!top], max_pts))
  n <- length(pvalues)
  if (is.null(n_total)) n_total <- n
  rank <- seq_len(n)
  a <- 0.5
  obs <- sort(-log10(pvalues))
  exp <- sort(-log10((rank - a) / (n_total + 1 - 2 * a)))
  ci_hi <- -log10(qbeta(0.025, rank, n_total - rank + 1))
  ci_lo <- -log10(qbeta(0.975, rank, n_total - rank + 1))

  ggplot(data.frame(exp=exp, obs=obs,
                     ci_hi=sort(ci_hi), ci_lo=sort(ci_lo)),
          aes(x=exp, y=obs)) +
    geom_ribbon(aes(ymin=ci_lo, ymax=ci_hi),
               fill="lightblue", alpha=0.4) +
    geom_point(size=0.5, alpha=0.5, color=color) +
    geom_abline(slope=1, intercept=0,
                color="red", linewidth=0.7) +
    labs(title = title,
          subtitle = sprintf("n = %s%s", format(n, big.mark=","),
                              if (n_total > n)
                                sprintf(" (of %s cis tests)", format(n_total,
big.mark=","))
                              else ""),
          x="Expected -log10(p)",
          y="Observed -log10(p)") +
    theme_minimal(base_size=11) +
    theme(plot.subtitle=element_text(size=9, color="grey40"))
}

```

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```

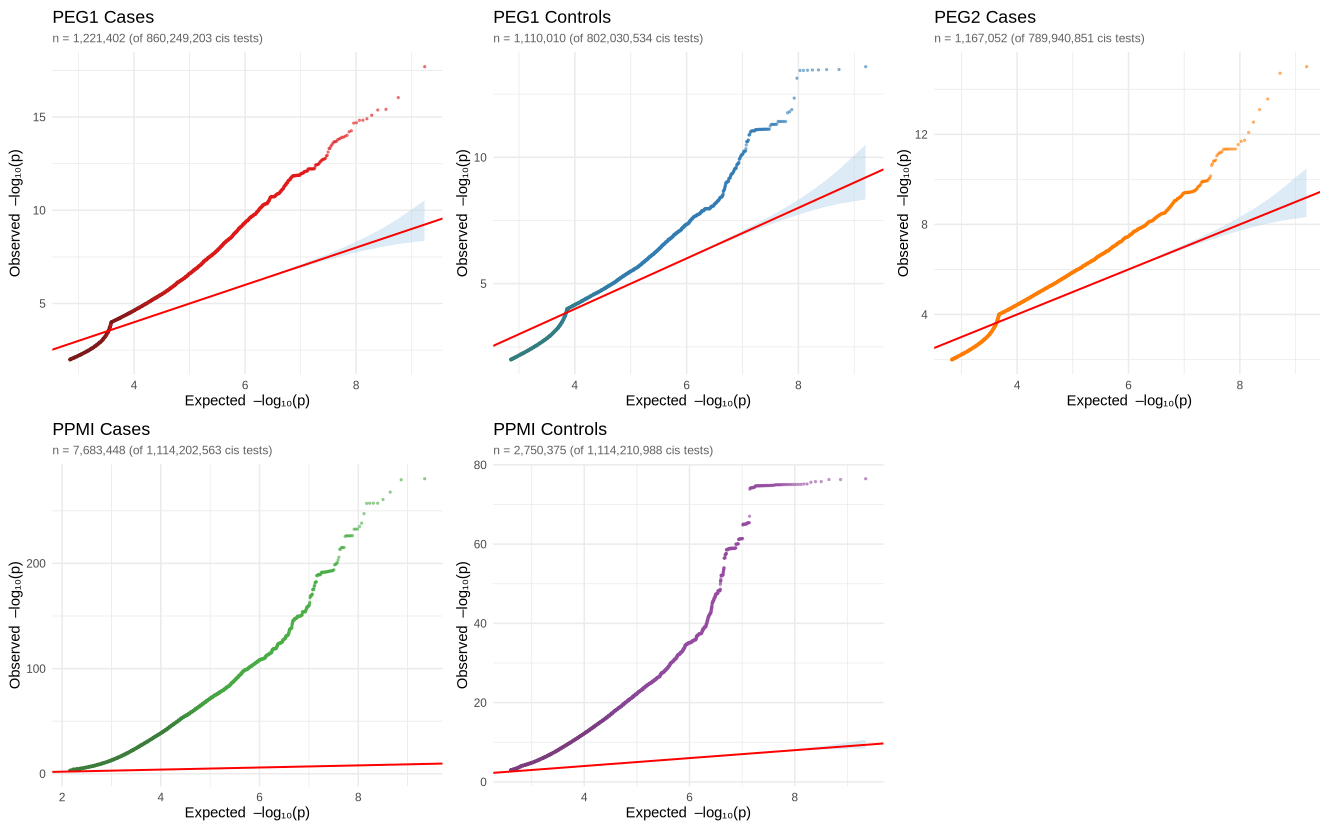
n_cis_tests_df <- read.csv("../results/META_MQTL/qc/n_cis_tests.csv")
n_cis_tests <- setNames(n_cis_tests_df$n_total, n_cis_tests_df$dataset)

qq_plots <- imap(mqtl_list, function(dat, nm)
  make_qq(dat$pvalue, DATASET_LABELS[nm], DATASET_COLORS[nm],
    n_total = n_cis_tests[[nm]]))

wrap_plots(qq_plots, ncol=3) +
  plot_annotation(
    title = "QQ Plots – Per-Dataset Cis mQTL Analyses",
    theme = theme(plot.title=element_text(size=15, face="bold"))
  )

```

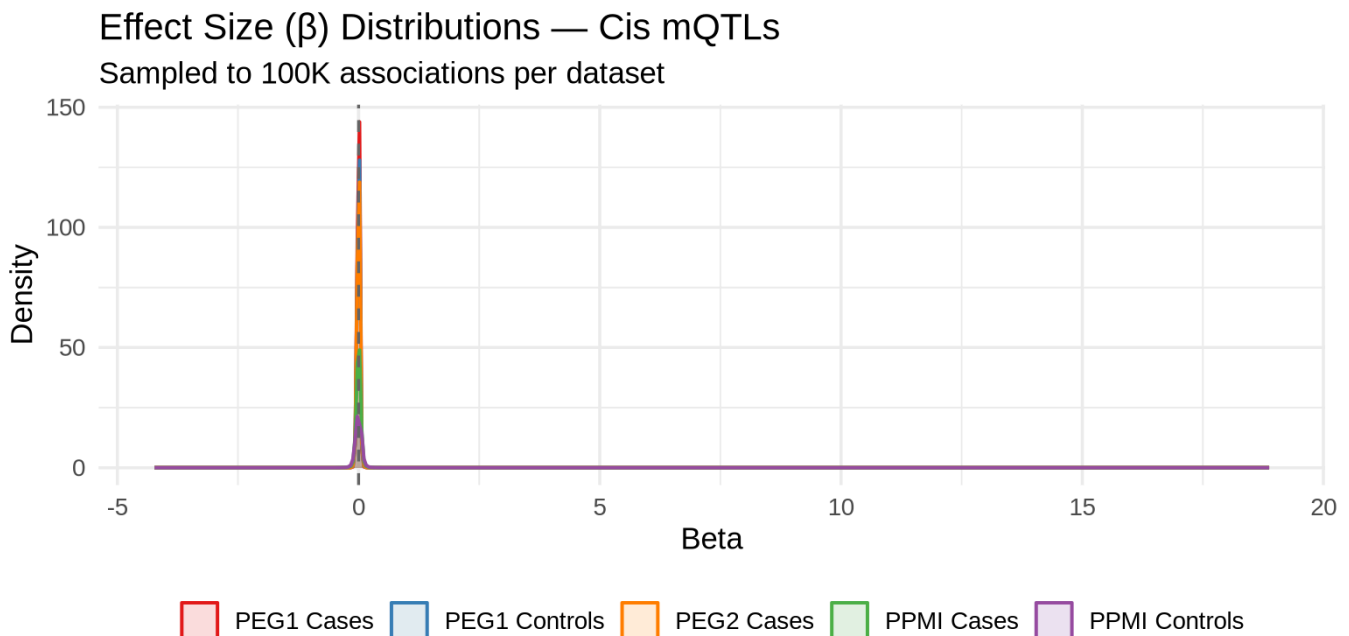
QQ Plots — Per-Dataset Cis mQTL Analyses



3.3 Effect Size Distributions

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```
imap_dfr(mqtl_list, function(dat, nm)
  dat %>%
    slice_sample(n=min(100000, nrow(dat))) %>%
    mutate(label=DATASET_LABELS[nm])
) %>%
  ggplot(aes(x=beta, color=dataset, fill=dataset)) +
  geom_density(alpha=0.15, linewidth=0.7) +
  scale_color_manual(values=DATASET_COLORS, labels=DATASET_LABELS) +
  scale_fill_manual(values=DATASET_COLORS, labels=DATASET_LABELS) +
  geom_vline(xintercept=0, linetype="dashed", color="grey40") +
  labs(title="Effect Size ( $\beta$ ) Distributions – Cis mQTLs",
       subtitle="Sampled to 100K associations per dataset",
       x="Beta", y="Density", color=NULL, fill=NULL) +
  theme_minimal(base_size=13) +
  theme(legend.position="bottom")
```



4. Meta-Analysis with r-meta

4.1 Strategy

We use **fixed-effects inverse-variance weighted (IVW) meta-analysis** via `metagen()` from the `meta` package. Fixed effects is appropriate here because we are combining results from cohorts drawn from the same biological population — we want to estimate a common true effect, not the distribution of effects across populations.

For each SNP-CpG pair tested in multiple datasets:

- **Effect size:** β from MatrixEQTL

- **Standard error:** $SE = |\beta| / t\text{-stat}$
- **Method:** IVW fixed effects (`method.tau="DL"` for heterogeneity)
- **Heterogeneity:** Cochran's Q and I^2 reported for significant hits

4.2 Run Meta-Analysis Per Group

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```

run_meta_group <- function(group_name, dataset_names) {
  cat(sprintf("\n=== %s meta-analysis ===\n", group_name))
  message(sprintf("\n=== %s meta-analysis ===\n", group_name))
  group_data <- mgtl_list[dataset_names]
  group_data <- Filter(Negate(is.null), group_data)

  if (length(group_data) == 0) {
    cat("No data available\n")
    return(NULL)
  }

  cat(sprintf("Datasets: %s\n", paste(names(group_data), collapse=", ")))
  message(sprintf("Datasets: %s\n", paste(names(group_data), collapse=", ")))
  # — Strict intersection —————
  —
  # Only meta-analyse pairs significant in ALL constituent datasets.
  # This is computationally tractable and focuses on robustly replicated sign
als.
  # Individual cohort thresholds are set permissively (pval_cis = 1e-3 for PE
G,
  # 1e-5 for PPMI) so the intersection captures pairs that clear the bar in b
oth.
  common_pairs <- Reduce(intersect, lapply(group_data, function(d) d$pair_i
d))
  cat(sprintf("SNP-CpG pairs in all datasets: %s\n",
    format(length(common_pairs), big.mark=",", "")))

  message(sprintf("SNP-CpG pairs in all datasets: %s\n",
    format(length(common_pairs), big.mark=",", "")))
  if (length(common_pairs) == 0) {
    cat("No common pairs – check SNP and probe intersection\n")
    return(NULL)
  }

  # Subset each dataset to common pairs and align
  aligned <- lapply(names(group_data), function(nm) {
    group_data[[nm]] %>%
      filter(pair_id %in% common_pairs) %>%
      select(pair_id, snp_id, cp_g_id, beta, se, pvalue) %>%
      rename_with(~paste0(., "_", nm), c(beta, se, pvalue, snp_id, cp_g_id))
  })

  # Inner join – all pairs present in all datasets by construction
  merged <- Reduce(function(a, b) inner_join(a, b, by="pair_id"), aligned)

  # Recover snp_id and cp_g_id from first dataset
  first_nm <- names(group_data)[1]
  merged <- merged %>%

```

```

    rename(snp_id = paste0("snp_id_", first_nm),
           cpg_id = paste0("cpg_id_", first_nm)) %>%
    select(-matches("^snp_id_|^cpg_id_"))

cat(sprintf("Pairs after alignment join: %s\n",
    format(nrow(merged), big.mark=",")))
message(sprintf("Pairs after alignment join: %s\n",
    format(nrow(merged), big.mark=",")))

beta_cols <- paste0("beta_", names(group_data))
se_cols    <- paste0("se_",   names(group_data))

cat("Running metagen()...\n")

message("Running metagen()...\n")

meta_results <- lapply(seq_len(nrow(merged)), function(i) {
  betas <- as.numeric(merged[i, beta_cols])
  ses    <- as.numeric(merged[i, se_cols])

  if (any(is.na(betas)) || any(is.na(ses)) || any(ses <= 0))
    return(NULL)

  mg <- tryCatch(
    metagen(TE=betas, seTE=ses,
            studlab   = names(group_data),
            method.tau = "DL",
            common     = TRUE,
            random     = FALSE),
    error = function(e) NULL
  )
  if (is.null(mg)) return(NULL)

  data.frame(
    pair_id   = merged$pair_id[i],
    snp_id    = merged$snp_id[i],
    cpg_id    = merged$cpg_id[i],
    meta_beta = mg$TE.common,
    meta_se   = mg$seTE.common,
    meta_pval = mg$pval.common,
    meta_z    = mg$zval.common,
    Q         = mg$Q,
    Q_pval    = mg$pval.Q,
    I2        = mg$I2,
    n_datasets = length(betas),
    stringsAsFactors = FALSE
  )
})

```

```

meta_df <- bind_rows(Filter(Negate(is.null), meta_results))

if (nrow(meta_df) == 0) {
  cat("No results produced\n")
  return(NULL)
}

# FDR correction
meta_df <- meta_df %>%
  arrange(meta_pval) %>%
  mutate(FDR_BH = p.adjust(meta_pval, method="BH"))

cat(sprintf("Meta-analysis complete: %s pairs\n",
  format(nrow(meta_df), big.mark=",")))
cat(sprintf("  FDR < 0.05: %s\n",
  format(sum(meta_df$FDR_BH < 0.05, na.rm=TRUE), big.mark=",")))
cat(sprintf("  FDR < 0.10: %s\n",
  format(sum(meta_df$FDR_BH < 0.10, na.rm=TRUE), big.mark=",")))
cat(sprintf("  p < 1e-8: %s\n",
  format(sum(meta_df$meta_pval < 1e-8, na.rm=TRUE), big.mark=",")))

# Save full results
out_file <- file.path(META_DIR,
  paste0(group_name, "_meta_results.csv.gz"))
fwrite(meta_df, out_file)
cat(sprintf("  Saved: %s\n", basename(out_file)))

invisible(meta_df)
}

# NOTE: metagen() is called once per SNP-CpG pair.
# For large datasets this can take 30-60 minutes.
# Results are cached – delete cache/ to rerun.
meta_results <- list()
for (group in names(META_GROUPS)) {
  meta_results[[group]] <- run_meta_group(group, META_GROUPS[[group]])
}

```

```

##
## === cases meta-analysis ===

```

```

## Datasets: pegl_cases, ppmi_cases

```

```

## SNP-CpG pairs in all datasets: 34,389

```

```

## Pairs after alignment join: 34,389

```

```
## Running metagen()...
```

```
## Meta-analysis complete: 34,389 pairs
##   FDR < 0.05: 22,407
##   FDR < 0.10: 23,344
##   p < 1e-8:   6,625
##   Saved: cases_meta_results.csv.gz
##
## === controls meta-analysis ===
```

```
## Datasets: pegl_controls, ppmi_controls
```

```
## SNP-CpG pairs in all datasets: 11,446
```

```
## Pairs after alignment join: 11,446
```

```
## Running metagen()...
```

```
## Meta-analysis complete: 11,446 pairs
##   FDR < 0.05: 4,470
##   FDR < 0.10: 4,715
##   p < 1e-8:   652
##   Saved: controls_meta_results.csv.gz
```

Hide

```
# — Sensitivity analysis: 3-way intersection including PEG2 cases ———
# PEG2 is cases-only (Wave 2). The 3-way PEG1 n PEG2 n PPMI intersection
# is very strict but identifies the most robustly replicated signals.
message("Running sensitivity analysis (PEG1 + PEG2 + PPMI cases 3-way)...")
sensitivity_results <- list()
for (group in names(SENSITIVITY_GROUPS)) {
  sensitivity_results[[group]] <- run_meta_group(
    group, SENSITIVITY_GROUPS[[group]])
}
```

```
##
## === cases_peg2_sensitivity meta-analysis ===
```

```
## Datasets: pegl_cases, peg2_cases, ppmi_cases
```

```
## SNP-CpG pairs in all datasets: 357
```

```
## Pairs after alignment join: 357
```

```
## Running metagen()...
```

```
## Meta-analysis complete: 357 pairs  
## FDR < 0.05: 213  
## FDR < 0.10: 254  
## p < 1e-8: 96  
## Saved: cases_peg2_sensitivity_meta_results.csv.gz
```

[Hide](#)

```
# — Flag primary cases hits also significant in PEG2 —————  
—  
if (!is.null(meta_results$cases) && nrow(meta_results$cases) > 0) {  
  peg2_pairs <- mql_list$peg2_cases$pair_id  
  meta_results$cases$in_peg2 <- meta_results$cases$pair_id %in% peg2_pairs  
  message(sprintf("Primary cases hits also significant in PEG2: %d / %d",  
    sum(meta_results$cases$in_peg2, na.rm=TRUE),  
    nrow(meta_results$cases)))  
}
```

5. Meta-Analysis Results

5.1 Summary

[Hide](#)

```

summary_df <- imap_dfr(meta_results, function(dat, group) {
  if (is.null(dat) || nrow(dat) == 0) return(NULL)
  data.frame(
    group      = str_to_title(group),
    n_pairs    = nrow(dat),
    pct_fdr05  = round(100 * mean(dat$FDR_BH < 0.05, na.rm=TRUE), 1),
    n_fdr05    = sum(dat$FDR_BH < 0.05, na.rm=TRUE),
    n_fdr10    = sum(dat$FDR_BH < 0.10, na.rm=TRUE),
    n_ple5     = sum(dat$meta_pval < 1e-5, na.rm=TRUE),
    n_ple8     = sum(dat$meta_pval < 1e-8, na.rm=TRUE),
    n_in_peg2  = if ("in_peg2" %in% colnames(dat))
                  sum(dat$in_peg2, na.rm=TRUE) else NA,
    pct_het    = round(100 * mean(dat$Q_pval < 0.05, na.rm=TRUE), 1),
    stringsAsFactors = FALSE
  )
})

if (is.null(summary_df) || nrow(summary_df) == 0) {
  cat("No meta-analysis results available yet – run Section 4.2 first.
")
} else {
  summary_df %>%
    mutate(across(c(n_pairs, n_fdr05, n_fdr10, n_ple5, n_ple8, n_in_peg2),
      ~format(.x, big.mark=","))) %>%
    knitr::kable(
      col.names = c("Group", "Pairs", "% FDR<0.05", "FDR<0.05",
        "FDR<0.10", "p<1e-5", "p<1e-8",
        "Also in PEG2", "% heterogeneous"),
      caption = paste("Meta-analysis summary (Cases: PEG1 × PPMI two-way;",
        "Controls: PEG1 × PPMI two-way).",
        "'Also in PEG2' = primary hits also significant in PEG2",
        "at pval_cis < 1e-3, providing three-cohort support.",
        "FDR_BH computed on meta-analysis p-values.")
    )
}

```

Meta-analysis summary (Cases: PEG1 × PPMI two-way; Controls: PEG1 × PPMI two-way). ‘Also in PEG2’ = primary hits also significant in PEG2 at pval_cis < 1e-3, providing three-cohort support. FDR_BH computed on meta-analysis p-values.

Group	Pairs	% FDR<0.05	FDR<0.05	FDR<0.10	p<1e-5	p<1e-8	Also in PEG2	% heterogeneous
Cases	34,389	65.2	22,407	23,344	17,941	6,625	357	61.6

Group	Pairs	% FDR<0.05	% FDR<0.05	% FDR<0.10	p<1e- 5	p<1e- 8	Also in PEG2	% heterogeneous
Controls	11,446	39.1	4,470	4,715	2,590	652	NA	83.3

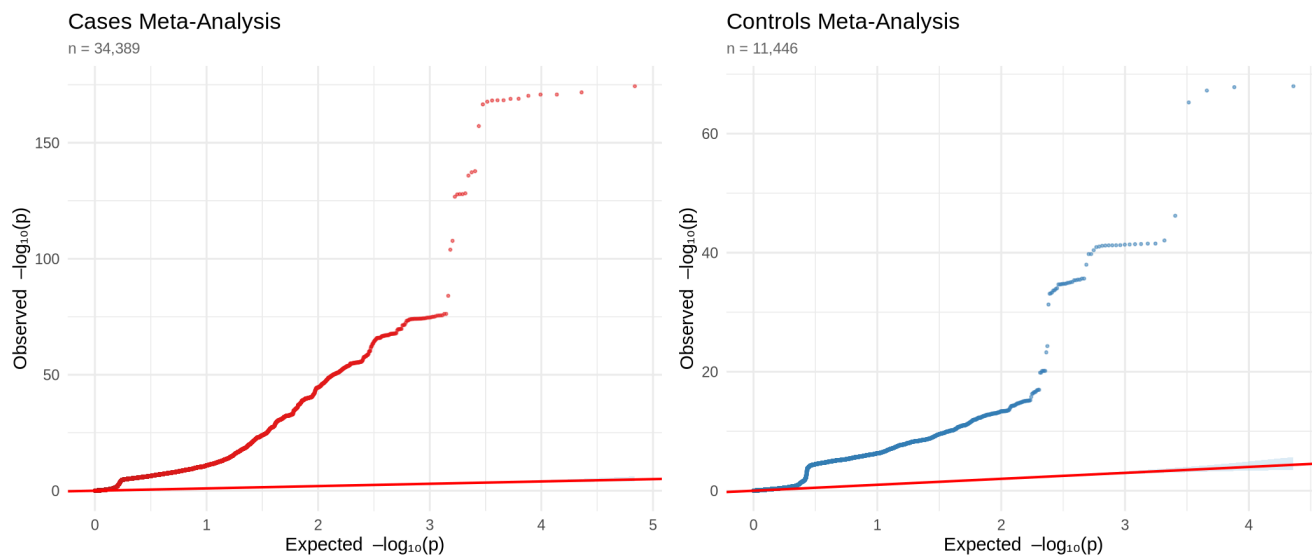
5.2 QQ Plots — Meta-Analysis

[Hide](#)

```
meta_qq <- imap(meta_results, function(dat, group) {
  if (is.null(dat)) return(NULL)
  make_qq(dat$meta_pval,
    title = sprintf("%s Meta-Analysis", str_to_title(group)),
    color = ifelse(group=="cases", "#E41A1C", "#377EB8"))
})
meta_qq <- Filter(Negate(is.null), meta_qq)

if (length(meta_qq) > 0)
  wrap_plots(meta_qq, ncol=2) +
    plot_annotation(
      title = "QQ Plots — Meta-Analysis Results",
      theme = theme(plot.title=element_text(size=15, face="bold"))
    )
```

QQ Plots — Meta-Analysis Results



5.3 Top Meta-mQTLs

[Hide](#)

```

fmt_pval <- function(x) {
  case_when(
    is.na(x)      ~ "NA",
    x < 1e-300    ~ "< 1e-300",
    x < 0.001     ~ formatC(x, format="e", digits=2),
    TRUE         ~ as.character(round(x, 4))
  )
}

for (group in names(meta_results)) {
  dat <- meta_results[[group]]
  if (is.null(dat) || nrow(dat) == 0) {
    cat(sprintf("\n### %s - no results\n", str_to_title(group)))
    next
  }

  cat(sprintf("\n### %s - Top 20\n", str_to_title(group)))

  top20 <- dat %>%
    arrange(meta_pval) %>%
    slice_head(n=20)

  # Pull per-study beta and p-value for each top pair
  datasets_in_group <- META_GROUPS[[group]]
  per_study <- lapply(datasets_in_group, function(nm) {
    mqt1_list[[nm]] %>%
      filter(pair_id %in% top20$pair_id) %>%
      select(pair_id, beta, pvalue) %>%
      rename_with(~paste0(nm, "_", .), c(beta, pvalue))
  })
  per_study_df <- Reduce(function(a, b)
    left_join(a, b, by="pair_id"), per_study)

  # Join per-study stats to top20
  top20 <- top20 %>%
    left_join(per_study_df, by="pair_id")

  # Build dynamic column list
  study_beta_cols <- paste0(datasets_in_group, "_beta")
  study_pval_cols <- paste0(datasets_in_group, "_pvalue")

  top20 %>%
    select(snp_id, cpg_id, meta_beta, meta_se,
           meta_pval, FDR_BH, I2, Q_pval,
           all_of(c(study_beta_cols, study_pval_cols))) %>%
    mutate(
      meta_beta = round(meta_beta, 4),
      meta_se   = round(meta_se, 4),

```

```

    meta_pval = fmt_pval(meta_pval),
    FDR_BH     = fmt_pval(FDR_BH),
    I2         = round(I2, 1),
    Q_pval     = fmt_pval(Q_pval),
    across(all_of(study_beta_cols), ~round(.x, 4)),
    across(all_of(study_pval_cols), ~fmt_pval(.x))
  ) %>%
knitr::kable(
  col.names = c(
    "SNP", "CpG", "Meta  $\beta$ ", "Meta SE",
    "Meta p", "FDR", "I2", "Q p",
    paste0(datasets_in_group, "  $\beta$ "),
    paste0(datasets_in_group, " p")
  ),
  caption = sprintf("Top 20 %s meta-mQTLs with per-study statistics",
                    group)
) %>%
print()
}

```

```

##
## ### Cases – Top 20
##
##
## Table: Top 20 cases meta-mQTLs with per-study statistics
##
## |SNP          |CpG          | Meta  $\beta$  | Meta SE|Meta p      |FDR          | I2 |Q p
## | pegl_cases  $\beta$  | ppmi_cases  $\beta$  | pegl_cases p | ppmi_cases p |
## |:-----|:-----|:-----|:-----|:-----|:-----|---:|:---
## |-----|:-----|:-----|:-----|:-----|:-----|
## |5:157079630 |cg07464716 | -0.1665| 0.0059|4.10e-175 |1.41e-170 | 1|1.37
e-38 | -0.0357| -0.2115|0.0024 |4.19e-78 |
## |5:157083004 |cg07464716 | -0.1661| 0.0059|1.84e-172 |3.17e-168 | 1|1.89
e-38 | -0.0357| -0.2114|0.0024 |1.28e-77 |
## |5:157086041 |cg07464716 | -0.1658| 0.0059|1.65e-171 |1.43e-167 | 1|2.35
e-38 | -0.0357| -0.2113|0.0024 |1.96e-77 |
## |5:157085814 |cg07464716 | -0.1658| 0.0059|1.66e-171 |1.43e-167 | 1|2.34
e-38 | -0.0357| -0.2113|0.0024 |1.96e-77 |
## |5:157087289 |cg07464716 | -0.1657| 0.0059|5.70e-171 |3.92e-167 | 1|2.53
e-38 | -0.0357| -0.2112|0.0024 |2.47e-77 |
## |5:157063194 |cg07464716 | -0.1633| 0.0059|1.03e-169 |5.68e-166 | 1|3.88
e-37 | -0.0355| -0.2072|0.0025 |6.65e-77 |
## |5:157064500 |cg07464716 | -0.1633| 0.0059|1.16e-169 |5.68e-166 | 1|3.87
e-37 | -0.0355| -0.2072|0.0024 |6.79e-77 |
## |5:157066471 |cg07464716 | -0.1632| 0.0059|4.61e-169 |1.79e-165 | 1|4.17
e-37 | -0.0355| -0.2072|0.0024 |8.81e-77 |
## |5:157067030 |cg07464716 | -0.1632| 0.0059|4.68e-169 |1.79e-165 | 1|4.16
e-37 | -0.0355| -0.2072|0.0024 |8.83e-77 |
## |5:157061963 |cg07464716 | -0.1632| 0.0059|5.88e-169 |2.02e-165 | 1|4.11
e-37 | -0.0356| -0.2072|0.0024 |9.19e-77 |
## |5:157072799 |cg07464716 | -0.1633| 0.0059|1.95e-168 |6.10e-165 | 1|3.17
e-37 | -0.0355| -0.2077|0.0025 |1.08e-76 |
## |5:157054725 |cg07464716 | -0.1629| 0.0059|2.82e-167 |8.07e-164 | 1|5.51
e-37 | -0.0356| -0.2071|0.0024 |1.94e-76 |
## |5:157105368 |cg07464716 | -0.1637| 0.0061|6.39e-158 |1.69e-154 | 1|2.21
e-37 | -0.0355| -0.2113|0.0028 |8.29e-75 |
## |5:157044476 |cg07464716 | -0.1563| 0.0062|1.76e-138 |4.33e-135 | 1|1.91
e-34 | -0.0358| -0.2047|0.0023 |2.10e-70 |
## |5:157043663 |cg07464716 | -0.1562| 0.0062|5.49e-138 |1.26e-134 | 1|2.14
e-34 | -0.0358| -0.2047|0.0023 |2.72e-70 |
## |5:157106636 |cg07464716 | -0.1575| 0.0063|1.42e-136 |3.05e-133 | 1|6.77
e-35 | -0.0354| -0.2074|0.0029 |4.04e-70 |
## |6:112168309 |cg26846943 | 0.1297| 0.0054|6.42e-129 |1.30e-125 | 1|7.13
e-37 | -0.0374| 0.1575|0.0091 |5.08e-69 |
## |6:112164441 |cg26846943 | 0.1297| 0.0054|1.40e-128 |2.54e-125 | 1|7.52
e-37 | -0.0374| 0.1575|0.0091 |6.07e-69 |
## |6:112164313 |cg26846943 | 0.1297| 0.0054|1.40e-128 |2.54e-125 | 1|7.49
e-37 | -0.0374| 0.1575|0.0091 |6.07e-69 |

```

```

## |6:112161786 |cg26846943 | 0.1296| 0.0054|1.79e-128 |3.08e-125 | 1|7.58
e-37 | -0.0374| 0.1575|0.0091 |6.41e-69 |
##
## ### Controls – Top 20
##
##
## Table: Top 20 controls meta-mQTLs with per-study statistics
##
## |SNP |CpG | Meta  $\beta$ | Meta SE|Meta p |FDR | I2|Q p
| pegl_controls  $\beta$ | ppmi_controls  $\beta$ |pegl_controls p |ppmi_controls p |
## |:-----|:-----|-----:|-----:|:-----|:-----|--:|:-----
---|-----:|-----:|:-----|:-----|
## |12:56358066 |cg00635560 | -0.1593| 0.0091|1.05e-68 |8.75e-65 | 1|4.60e-
40 | 0.0473| -0.2296|0.0094 |4.83e-29 |
## |12:56359085 |cg00635560 | -0.1600| 0.0091|1.53e-68 |8.75e-65 | 1|1.86e-
40 | 0.0473| -0.2315|0.0094 |4.57e-29 |
## |12:56359878 |cg00635560 | -0.1589| 0.0091|5.69e-68 |2.17e-64 | 1|1.36e-
41 | 0.0504| -0.2315|0.0055 |4.20e-29 |
## |12:56356559 |cg00635560 | -0.1578| 0.0092|5.62e-66 |1.61e-62 | 1|8.91e-
40 | 0.0473| -0.2296|0.0094 |1.01e-28 |
## |9:124240842 |cg13607226 | 0.0772| 0.0054|6.25e-47 |1.43e-43 | 1|1.49e-
39 | -0.0260| 0.1255|0.0068 |2.01e-26 |
## |12:56363666 |cg00635560 | -0.1376| 0.0100|8.92e-43 |1.70e-39 | 1|7.60e-
35 | 0.0474| -0.2202|0.0094 |3.15e-25 |
## |9:124248449 |cg13607226 | 0.0743| 0.0055|2.96e-42 |4.22e-39 | 1|3.95e-
38 | -0.0271| 0.1232|0.005 |1.39e-25 |
## |9:124250678 |cg13607226 | 0.0744| 0.0055|3.03e-42 |4.22e-39 | 1|3.36e-
38 | -0.0271| 0.1234|0.005 |1.36e-25 |
## |9:124244677 |cg13607226 | 0.0743| 0.0055|3.51e-42 |4.22e-39 | 1|4.71e-
38 | -0.0271| 0.1231|0.0051 |1.46e-25 |
## |9:124244972 |cg13607226 | 0.0742| 0.0055|3.74e-42 |4.22e-39 | 1|4.37e-
38 | -0.0271| 0.1231|0.005 |1.46e-25 |
## |9:124252032 |cg13607226 | 0.0743| 0.0055|4.19e-42 |4.22e-39 | 1|3.38e-
38 | -0.0271| 0.1235|0.005 |1.43e-25 |
## |9:124256979 |cg13607226 | 0.0743| 0.0055|4.45e-42 |4.22e-39 | 1|2.95e-
38 | -0.0271| 0.1236|0.005 |1.41e-25 |
## |9:124259645 |cg13607226 | 0.0743| 0.0055|5.39e-42 |4.22e-39 | 1|2.97e-
38 | -0.0271| 0.1238|0.0051 |1.45e-25 |
## |9:124256843 |cg13607226 | 0.0743| 0.0055|5.68e-42 |4.22e-39 | 1|3.11e-
38 | -0.0271| 0.1236|0.005 |1.47e-25 |
## |9:124255690 |cg13607226 | 0.0742| 0.0055|5.83e-42 |4.22e-39 | 1|3.19e-
38 | -0.0271| 0.1236|0.005 |1.48e-25 |
## |9:124242638 |cg13607226 | 0.0741| 0.0055|5.90e-42 |4.22e-39 | 1|4.87e-
38 | -0.0271| 0.1231|0.005 |1.58e-25 |
## |9:124241603 |cg13607226 | 0.0741| 0.0055|6.36e-42 |4.28e-39 | 1|4.96e-
38 | -0.0271| 0.1231|0.005 |1.60e-25 |
## |9:124253878 |cg13607226 | 0.0742| 0.0055|7.08e-42 |4.50e-39 | 1|3.82e-
38 | -0.0271| 0.1235|0.005 |1.56e-25 |
## |9:124260929 |cg13607226 | 0.0742| 0.0055|9.28e-42 |5.59e-39 | 1|2.98e-

```

38		-0.0271		0.1238		0.005		1.57e-25	
##		9:124261232		cg13607226		0.0742		0.0055	
38		-0.0271		0.1238		0.005		1.65e-25	

5.4 Cases vs Controls Comparison

For mQTL pairs significant in both meta-analyses, compare effect sizes:

Hide

```

if (!is.null(meta_results$cases) && !is.null(meta_results$controls)) {

  shared <- inner_join(
    meta_results$cases %>% select(pair_id, snp_id, cp_g_id,
                                beta_cases=meta_beta,
                                pval_cases=meta_pval,
                                fdr_cases=FDR_BH),
    meta_results$controls %>% select(pair_id,
                                    beta_ctrls=meta_beta,
                                    pval_ctrls=meta_pval,
                                    fdr_ctrls=FDR_BH),
    by="pair_id"
  )

  cat(sprintf("Pairs tested in both groups: %s\n",
              format(nrow(shared), big.mark=",")))

  sig_either <- shared %>%
    filter(fdr_cases < 0.05 | fdr_ctrls < 0.05) %>%
    mutate(
      sig_group = case_when(
        fdr_cases < 0.05 & fdr_ctrls < 0.05 ~ "Both",
        fdr_cases < 0.05 ~ "Cases only",
        fdr_ctrls < 0.05 ~ "Controls only",
        TRUE ~ "Neither"
      )
    )

  cat(sprintf("FDR<0.05 in either group: %s\n",
              format(nrow(sig_either), big.mark=",")))
  print(table(sig_either$sig_group))

  ggplot(sig_either,
        aes(x=beta_cases, y=beta_ctrls, color=sig_group)) +
    geom_point(alpha=0.5, size=1.2) +
    geom_abline(slope=1, intercept=0,
                linetype="dashed", color="grey40") +
    geom_hline(yintercept=0, color="grey70") +
    geom_vline(xintercept=0, color="grey70") +
    scale_color_manual(values=c(
      "Both" = "#4DAF4A",
      "Cases only" = "#E41A1C",
      "Controls only" = "#377EB8"
    )) +
    labs(title = "Cases vs Controls Meta-mQTL Effect Sizes",
         subtitle= "FDR < 0.05 in at least one group",
         x = " $\beta$  (Cases meta-analysis)",
         y = " $\beta$  (Controls meta-analysis)",

```

```

    color = NULL) +
  theme_minimal(base_size=13) +
  theme(legend.position="bottom")
}

```

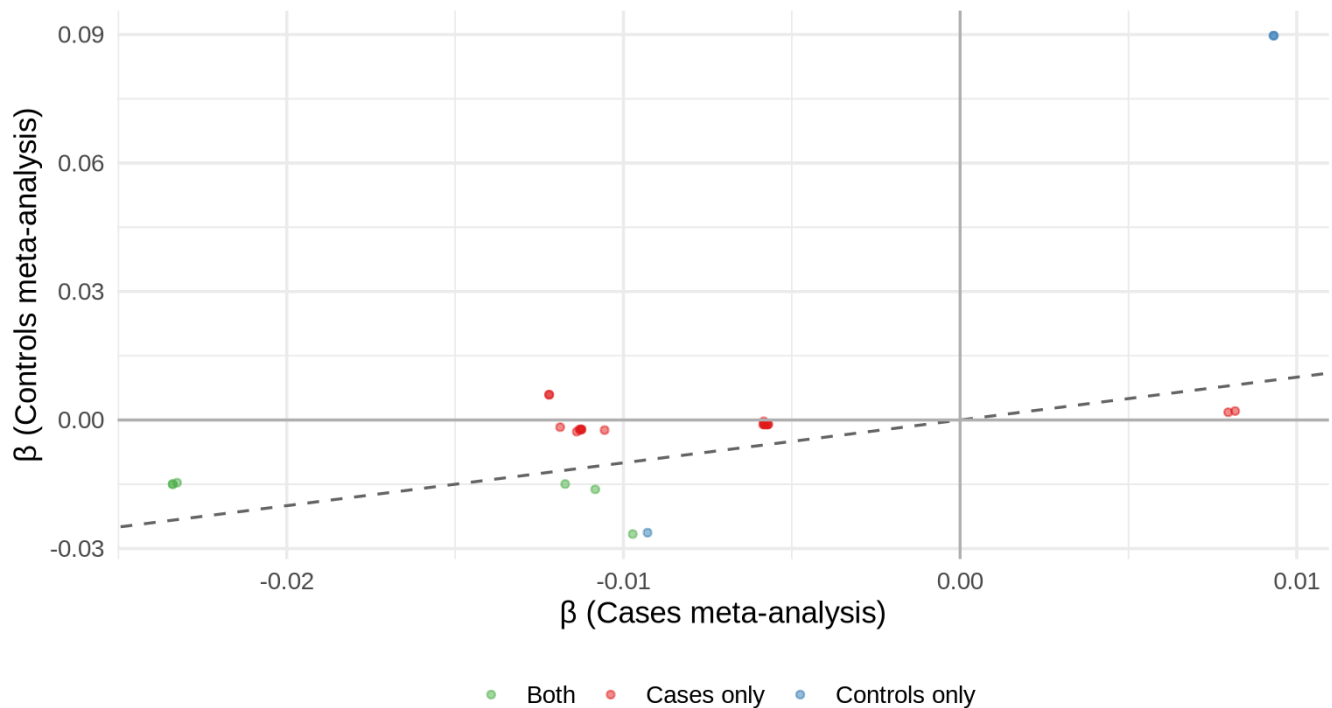
```

## Pairs tested in both groups: 40
## FDR<0.05 in either group: 28
##
##           Both    Cases only Controls only
##           6      19          3

```

Cases vs Controls Meta-mQTL Effect Sizes

FDR < 0.05 in at least one group



5.4b Volcano Plots

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```

volcano_plots <- imap(meta_results, function(dat, group) {
  if (is.null(dat) || nrow(dat) == 0) return(NULL)

  # Significance thresholds
  fdr_thresh <- dat %>% filter(FDR_BH < 0.05) %>%
    pull(meta_pval) %>% max(na.rm=TRUE)

  dat_plot <- dat %>%
    mutate(
      neg_log10_p = -log10(meta_pval),
      sig = case_when(
        FDR_BH < 0.05 & meta_beta > 0 ~ "FDR<0.05 (positive)",
        FDR_BH < 0.05 & meta_beta < 0 ~ "FDR<0.05 (negative)",
        TRUE ~ "Not significant"
      )
    )

  n_sig <- sum(dat_plot$sig != "Not significant", na.rm=TRUE)
  n_pos <- sum(dat_plot$sig == "FDR<0.05 (positive)", na.rm=TRUE)
  n_neg <- sum(dat_plot$sig == "FDR<0.05 (negative)", na.rm=TRUE)

  sig_colors <- c(
    "FDR<0.05 (positive)" = "#E41A1C",
    "FDR<0.05 (negative)" = "#377EB8",
    "Not significant" = "grey70"
  )

  ggplot(dat_plot, aes(x=meta_beta, y=neg_log10_p, color=sig)) +
    geom_point(size=0.7, alpha=0.5) +
    geom_hline(yintercept=-log10(fdr_thresh),
              linetype="dashed", color="grey30", linewidth=0.5) +
    geom_vline(xintercept=0,
              linetype="dashed", color="grey30", linewidth=0.5) +
    scale_color_manual(values=sig_colors) +
    annotate("text", x=Inf, y=-log10(fdr_thresh),
            label="FDR=0.05", hjust=1.1, vjust=-0.5,
            size=3, color="grey30") +
    annotate("text", x=Inf, y=Inf,
            label=sprintf("FDR<0.05: %s\n(+) %s  (-) %s",
                          format(n_sig, big.mark=","),
                          format(n_pos, big.mark=","),
                          format(n_neg, big.mark=",")),
            hjust=1.05, vjust=1.3, size=3.2, color="grey20") +
    labs(
      title = sprintf("%s Meta-Analysis Volcano Plot",
                      str_to_title(group)),
      subtitle = sprintf("%s total pairs",
                          format(nrow(dat), big.mark=",")),
    )

```

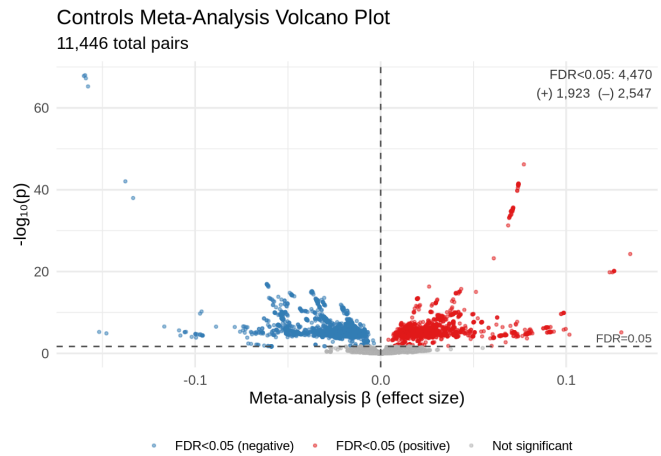
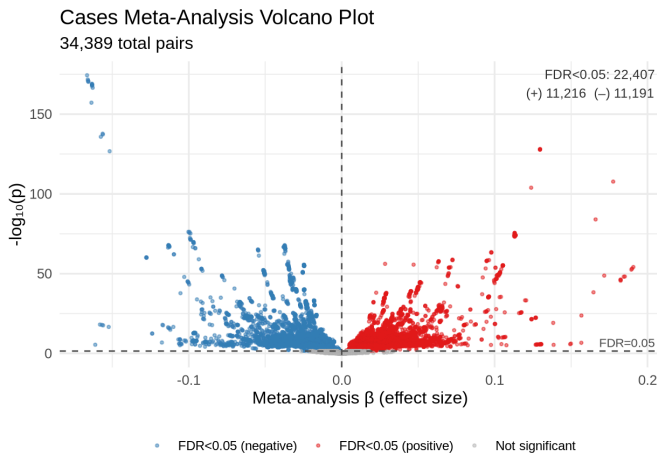
```

x      = "Meta-analysis  $\beta$  (effect size)",
y      = "-log10(p)",
color  = NULL
) +
theme_minimal(base_size=12) +
theme(legend.position="bottom",
      legend.text=element_text(size=9))
})

volcano_plots <- Filter(Negate(is.null), volcano_plots)
if (length(volcano_plots) > 0)
  wrap_plots(volcano_plots, ncol=2) +
    plot_annotation(
      title = "Volcano Plots — Meta-Analysis Results",
      theme = theme(plot.title=element_text(size=15, face="bold"))
    )

```

Volcano Plots — Meta-Analysis Results



5.5 Heterogeneity Summary

Hide

```

for (group in names(meta_results)) {
  dat <- meta_results[[group]]
  if (is.null(dat)) next

  sig <- dat %>% filter(FDR_BH < 0.05)
  if (nrow(sig) == 0) next

  cat(sprintf("\n%s – heterogeneity among FDR<0.05 hits:\n",
    str_to_title(group)))
  cat(sprintf("  Median I2:                %.1f%%\n",
    median(sig$I2, na.rm=TRUE)))
  cat(sprintf("  %% with Q p<0.05:          %.1f%%\n",
    100 * mean(sig$Q_pval < 0.05, na.rm=TRUE)))
  cat(sprintf("  %% with I2 > 50%%:          %.1f%%\n",
    100 * mean(sig$I2 > 50, na.rm=TRUE)))

  # I2 distribution plot
  p <- ggplot(sig, aes(x=I2)) +
    geom_histogram(bins=40, fill="steelblue", color="white") +
    geom_vline(xintercept=c(25,50,75),
      linetype="dashed", color="firebrick", alpha=0.6) +
    annotate("text", x=c(25,50,75), y=Inf,
      label=c("Low", "Moderate", "High"),
      vjust=2, hjust=-0.1, size=3, color="firebrick") +
    labs(title = sprintf("%s: I2 Distribution (FDR<0.05 hits)",
      str_to_title(group)),
      x="I2 (%)", y="Count") +
    theme_minimal(base_size=12)
  print(p)
}

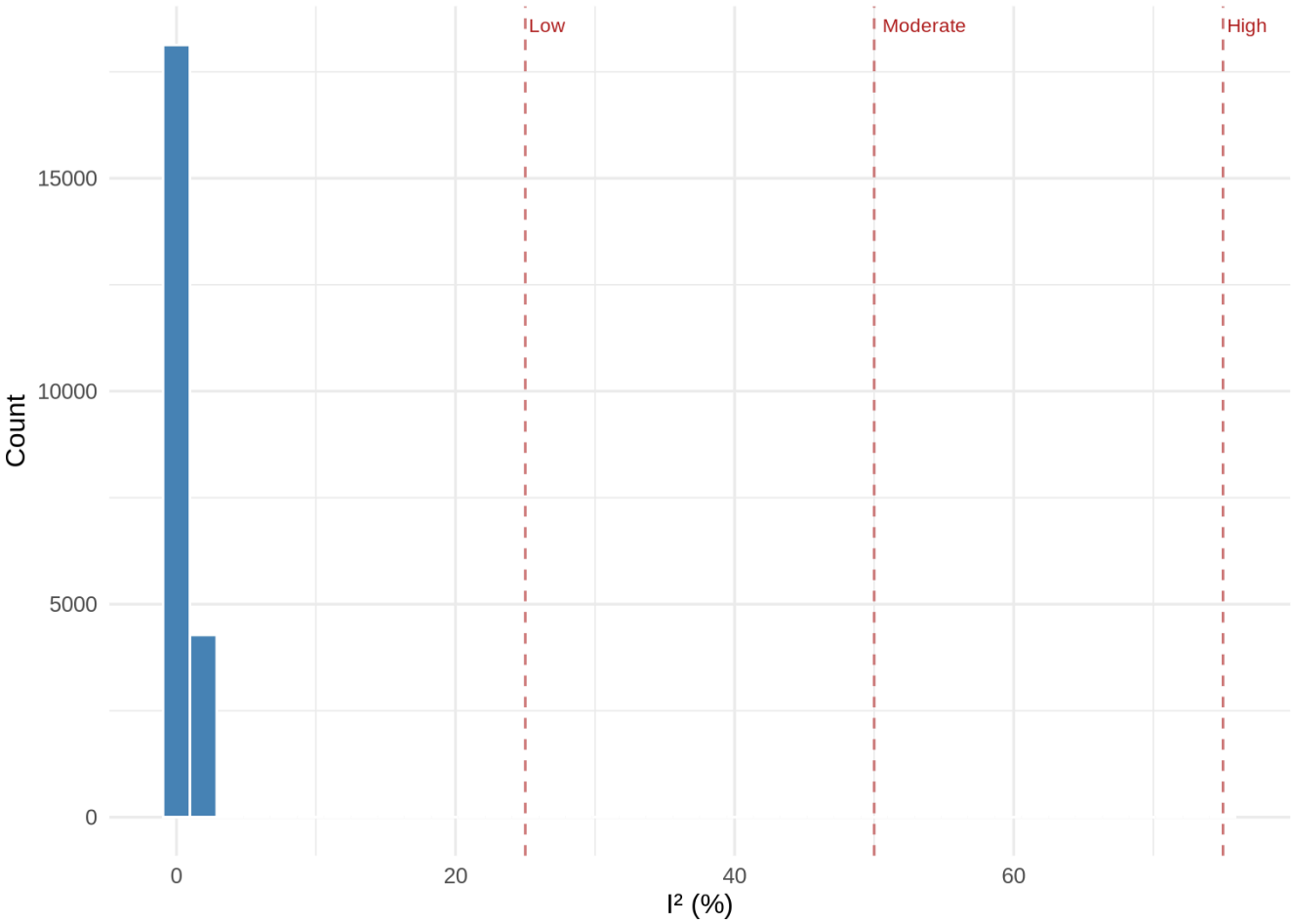
```

```

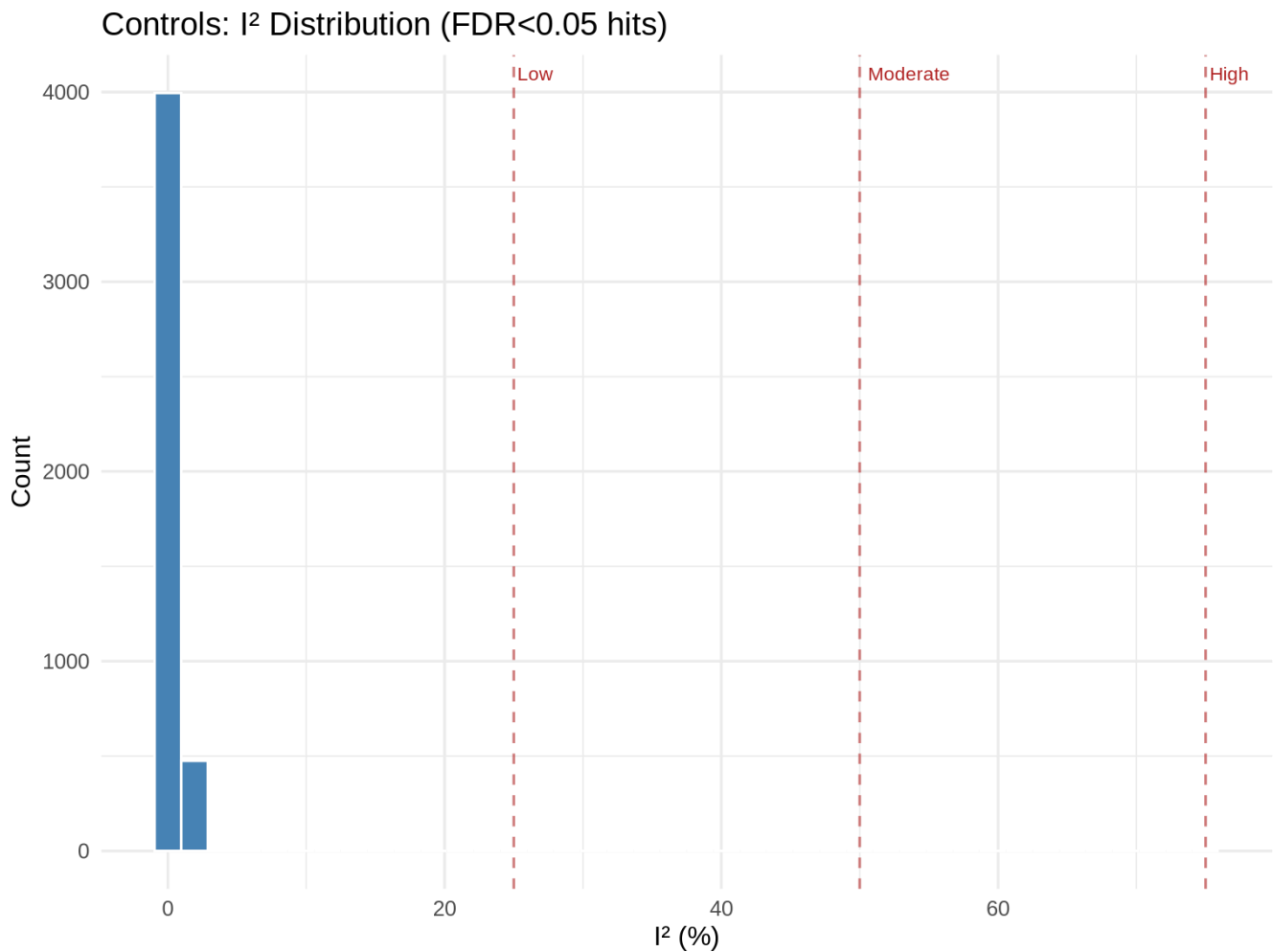
##
## Cases – heterogeneity among FDR<0.05 hits:
##   Median I2:                0.6%
##   % with Q p<0.05:          41.0%
##   % with I2 > 50%:          0.0%

```

Cases: I² Distribution (FDR<0.05 hits)



```
##
## Controls – heterogeneity among FDR<0.05 hits:
##   Median I2:           0.8%
##   % with Q p<0.05:      57.2%
##   % with I2 > 50%:     0.0%
```



6. Save Significant Results

Hide

```
for (group in names(meta_results)) {
  dat <- meta_results[[group]]
  if (is.null(dat)) next

  sig <- dat %>% filter(FDR_BH < 0.05)
  out <- file.path(META_DIR,
                   paste0(group, "_sig_fdr05.csv"))
  write.csv(sig, out, row.names=FALSE, quote=FALSE)
  cat(sprintf("%s: %s significant pairs (FDR<0.05) saved to %s\n",
              str_to_title(group),
              format(nrow(sig), big.mark=","),
              basename(out)))
}
```

```
## Cases: 22,407 significant pairs (FDR<0.05) saved to cases_sig_fdr05.csv
## Controls: 4,470 significant pairs (FDR<0.05) saved to controls_sig_fdr05.csv
```

7. Session Info

[Hide](#)

```
sessionInfo()
```

```

## R version 4.5.3 (2026-03-11)
## Platform: x86_64-conda-linux-gnu
## Running under: Ubuntu 24.04.4 LTS
##
## Matrix products: default
## BLAS: /home/garyc/miniconda3/envs/pd-qt1/lib/libblas.so.3.11.0
## LAPACK: /home/garyc/miniconda3/envs/pd-qt1/lib/liblapack.so.3.11.0 LAPACK
version 3.11.0
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8 LC_NUMERIC=C
## [3] LC_TIME=en_US.UTF-8 LC_COLLATE=en_US.UTF-8
## [5] LC_MONETARY=en_US.UTF-8 LC_MESSAGES=en_US.UTF-8
## [7] LC_PAPER=en_US.UTF-8 LC_NAME=C
## [9] LC_ADDRESS=C LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: America/Los_Angeles
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
## [1] scales_1.4.0 patchwork_1.3.2 meta_8.5-0 metabook_0.2-0
## [5] data.table_1.17.8 lubridate_1.9.5 forcats_1.0.1 stringr_1.6.0
## [9] dplyr_1.2.1 purrr_1.2.2 readr_2.2.0 tidyr_1.3.2
## [13] tibble_3.3.1 ggplot2_4.0.3 tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] sass_0.4.10 generics_0.1.4 xml2_1.5.2
## [4] stringi_1.8.7 lattice_0.22-9 lme4_2.0-6
## [7] hms_1.1.4 digest_0.6.39 magrittr_2.0.5
## [10] evaluate_1.0.5 grid_4.5.3 timechange_0.4.0
## [13] RColorBrewer_1.1-3 CompQuadForm_1.4.4 fastmap_1.2.0
## [16] Matrix_1.7-5 jsonlite_2.0.0 codetools_0.2-20
## [19] numDeriv_2016.8-1.1 jquerylib_0.1.4 reformulas_0.4.4
## [22] Rdpack_2.6.6 cli_3.6.6 rlang_1.2.0
## [25] rbibutils_2.4.1 splines_4.5.3 withr_3.0.2
## [28] cachem_1.1.0 yaml_2.3.12 ote1_0.2.0
## [31] tools_4.5.3 tzdb_0.5.0 nloptr_2.2.1
## [34] metafor_5.0-1 minqa_1.2.8 mathjaxr_2.0-0
## [37] boot_1.3-32 vctrs_0.7.3 R6_2.6.1
## [40] lifecycle_1.0.5 MASS_7.3-65 pkgconfig_2.0.3
## [43] pillar_1.11.1 bslib_0.11.0 gtable_0.3.6
## [46] Rcpp_1.1.1-1.1 glue_1.8.1 xfun_0.58
## [49] tidyselect_1.2.1 knitr_1.51 farver_2.1.2
## [52] nlme_3.1-169 htmltools_0.5.9 labeling_0.4.3

```

```
## [55] rmarkdown_2.31      compiler_4.5.3      S7_0.2.2
## [58] metadat_1.6-0
```